

# SaiFE Findings Summary

Safe AI Framework for Speed-Limit Evaluation

**14,711**

usable segments scored

**6,729**

priority segments scoring 50+

**131**

critical-review segments

**250**

time-based pilot candidates

## What Problem SaiFE Solves

SaiFE helps transport agencies identify road segments where posted speed limits, actual operating speeds, road context, and geometry may create elevated safety concerns, especially for pedestrians, cyclists, and powered two-wheeler users. It is a transparent screening tool for deciding where governments should investigate first.

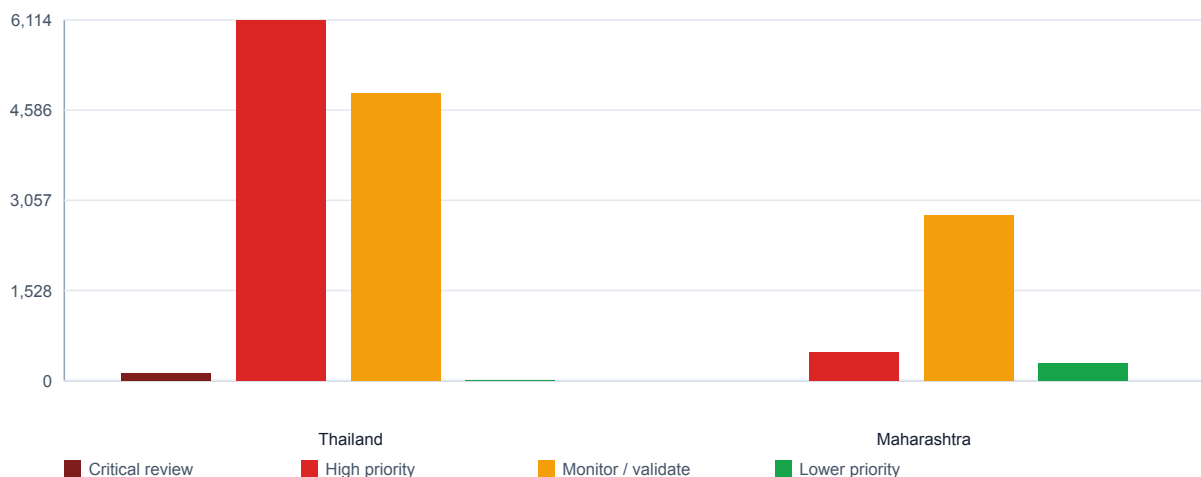
## Method in Brief

The model uses the supplied Thailand and Maharashtra GeoJSON datasets. Each valid segment receives a 0-100 Speed Safety Score from six components: posted limit alignment, operating speed behavior, exposure proxy, vulnerable-user context, geometry, and data-quality review. Segments scoring 50 or above are treated as priority speed-unsafe segments for review.

## Core Findings

Thailand produced 131 Critical-review segments and 6,114 High-priority segments. Maharashtra produced 484 High-priority segments. The strongest signals occur where high observed speeds combine with urban or major-road context, geometry concerns, and exposure proxies.

Priority classes by country



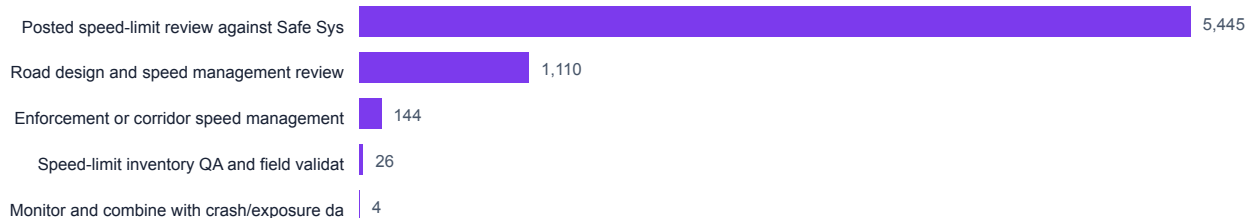
## Segments by Speed Safety Score band



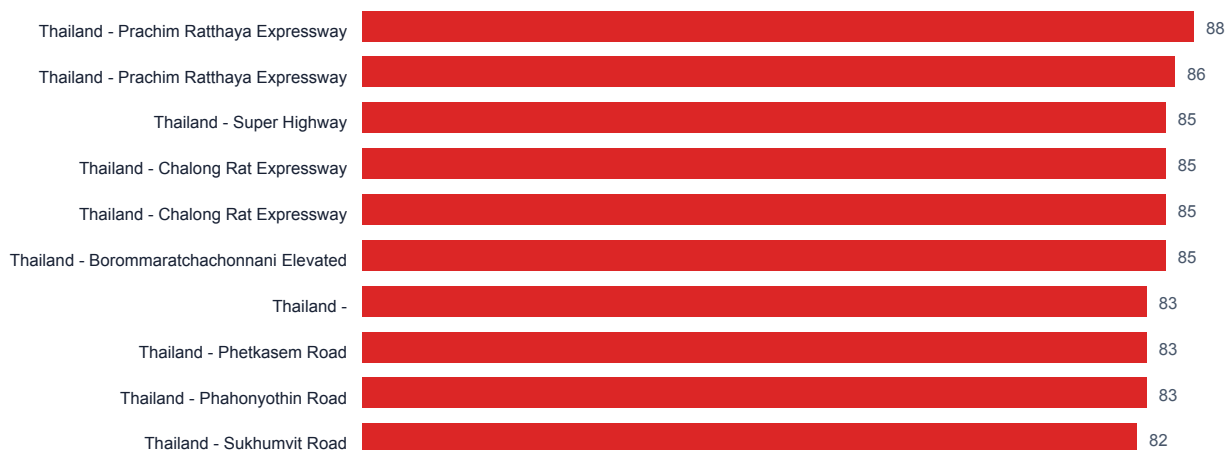
## Policy-Relevant Outputs

SaiFE does not only rank locations. It assigns action categories that a ministry can use for speed-limit review, road-design review, corridor speed management, enforcement planning, QA, or monitoring.

### Recommended action categories for priority segments



### Top scored segments



### Top Scored Segments

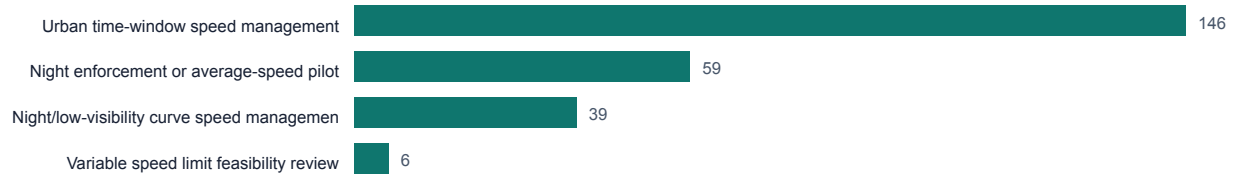
| Country  | Road                                   | Class    | Land use | Limit | F85          | Score | Recommended action                      |
|----------|----------------------------------------|----------|----------|-------|--------------|-------|-----------------------------------------|
| Thailand | Prachin Ratthaya Expressway            | motorway | URBAN    | 80.0  | 114.5        | 88    | Road design and speed management review |
| Thailand | Prachin Ratthaya Expressway            | motorway | URBAN    | 80.0  | 114.83333333 | 86    | Road design and speed management review |
| Thailand | Super Highway                          | trunk    | URBAN    | 80.0  | 95.5         | 85    | Road design and speed management review |
| Thailand | Chalong Rat Expressway                 | motorway | URBAN    | 80.0  | 108.66666666 | 85    | Road design and speed management review |
| Thailand | Chalong Rat Expressway                 | motorway | URBAN    | 80.0  | 107.11111111 | 85    | Road design and speed management review |
| Thailand | Borommaratchachonnani Elevated Highway | primary  | URBAN    | 80.0  | 99.85714285  | 85    | Road design and speed management review |
| Thailand |                                        | trunk    | URBAN    | 90.0  | 113.33333333 | 83    | Road design and speed management review |
| Thailand | Phetkasem Road                         | trunk    | URBAN    | 80.0  | 97.2         | 83    | Road design and speed management review |

| Country  | Road                  | Class | Land use | Limit | F85   | Score | Recommended action                      |
|----------|-----------------------|-------|----------|-------|-------|-------|-----------------------------------------|
| Thailand | Phahonyothin Road     | trunk | URBAN    | 80.0  | 108.0 | 83    | Road design and speed management review |
| Thailand | Sukhumvit Road        | trunk | URBAN    | 80.0  | 95.0  | 82    | Road design and speed management review |
| Thailand | Yantarakit Koson Road | trunk | URBAN    | 80.0  | 103.0 | 82    | Road design and speed management review |
| Thailand | Phet Kasem Road       | trunk | URBAN    | 80.0  | 91.5  | 82    | Road design and speed management review |

## Time-Based Speed Management

The provided GeoJSON files do not contain timestamp, hour, or day/night fields, so SaiFE does not claim to prove day/night speeding patterns. Instead, it identifies where segment-hour data should be collected first for variable speed limits, night enforcement, school-zone windows, or dynamic speed-management pilots.

### Time-based pilot candidate types



### First-Wave Time-Based Pilot Candidates

| Country  | Road                                   | Class    | Score | Pilot type                                  | Pilot score |
|----------|----------------------------------------|----------|-------|---------------------------------------------|-------------|
| Thailand | Prachin Ratthaya Expressway            | motorway | 88    | Night/low-visibility curve speed management | 116.59      |
| Thailand | Prachin Ratthaya Expressway            | motorway | 86    | Night/low-visibility curve speed management | 114.55      |
| Thailand | Chalong Rat Expressway                 | motorway | 85    | Night/low-visibility curve speed management | 111.48      |
| Thailand | Chalong Rat Expressway                 | motorway | 85    | Night/low-visibility curve speed management | 110.41      |
| Thailand | Udon Ratthaya Expressway               | motorway | 79    | Night enforcement or average-speed pilot    | 107.54      |
| Thailand | Borommaratchachonnani Elevated Highway | primary  | 85    | Urban time-window speed management          | 106.43      |
| Thailand | Buraphawithi Expressway                | motorway | 79    | Night enforcement or average-speed pilot    | 106.25      |
| Thailand | Phahonyothin Road                      | trunk    | 83    | Urban time-window speed management          | 104.85      |

## How a Ministry Could Use This

- Prioritize corridors for speed-limit review and Safe System alignment checks.
- Select locations for average-speed enforcement, fixed cameras, or speed feedback signs.
- Identify curved or high-speed roads where engineering treatment may be needed.
- Target field validation and speed-limit inventory QA.
- Choose first-wave corridors for segment-hour day/night speed analysis.

## Validation and Limitations

SaiFE is a screening and prioritization model, not a final crash-prediction model. The supplied data does not include crash outcomes, traffic volumes, vulnerable-user exposure, or time-of-day speed records. Final investment decisions should join the score with crash data, traffic counts, field audits, and local engineering review.

## Replication

The method can be replicated in other countries using common road data: segment geometry, posted speed limit, road class, land use, observed speed statistics, and sample count. Countries with less data can start with posted limits, geometry, road class, and urban/rural context, then add speed observations and crash data as they become available.

**Repository:** <https://github.com/tan-amos/adb-ai-for-safer-roads>

**Interactive map:** <https://tan-amos.github.io/adb-ai-for-safer-roads/>